Temporal Retiming Attacks on Hybrid Quantum Spiking Classifiers

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Abstract. Event timing is an adversarial input dimension, but attack comparisons depend on both the feasible retiming and the computation allocated to the search. We study this issue in QSNN-v3, an implemented hybrid of a classical Conv/LIF frontend and an analytically simulated eight-qubit probability readout. The model achieves $97.27\% \pm 0.21\%$ validation accuracy across five seeds conditional on shared pretraining and split. On 100 jointly clean-correct seed-42 validation examples, timestamp PGD succeeds on three examples at each tested radius. Derivative-free margin search produces zero to two additional successes depending on candidate, forward-pass or elapsed-time matching; no paired comparison establishes significant superiority. Separate classical packet-grid controls show that similar clean accuracy can accompany sharply different outcomes under one attack recipe, without isolating a representation or quantum effect. The evidence supports protocol-specific temporal vulnerability; official-test hybrid robustness and a quantum-specific mechanism remain unmeasured.

Keywords: event-driven classification; adversarial retiming; hybrid quantum classifiers; spiking neural networks; matched attack evaluation.

1 Introduction

Event-driven classifiers process not only where an event occurs but when it occurs. Retiming can change a prediction while preserving event identities, spatial coordinates, polarity and total count. Surrogate-gradient learning makes spiking dynamics trainable [1], while spike-retiming attacks make timing an explicit adversarial input dimension [2]. An evaluation must therefore specify both the admissible temporal changes and the computation available to find them.

We investigate end-to-end temporal vulnerability in an implemented hybrid classifier, emphasizing how computational matching changes attack comparisons and why results from separate temporal representations require separate interpretation. Our primary target, QSNN-v3, maps events through a classical Conv/LIF frontend, learned angles, an eight-qubit circuit and an affine probability readout. The central question is whether conclusions about its observed attack outcomes change with the definition of matched computation. Candidate count, forward-pass count and elapsed time measure different resources, particularly when only one attack uses gradients.

On the same 100 joint clean-correct validation examples, timestamp PGD causes three hybrid failures at each tested radius. TEMP-DRIFT-v2, a derivative-free margin search, causes 3–5 depending on radius and computational control. The largest numerical difference occurs under elapsed-time matching; no paired test establishes superiority. A separate four-model classical packet-grid study supplies context: similar clean accuracy need not imply similar attack outcomes. Differences in victims, populations, checkpoints and budget units prevent a causal representation ablation or a pooled robustness ranking across these studies.

The evaluation concerns the implemented composition. Quantum-classifier robustness can depend on encoding, model structure and adversarial access [3–5]; here the perturbation enters before the classical frontend. The following concrete outputs organize the paper:

1. **An explicit hybrid attack interface.** We specify temporal aggregation, Conv/LIF features and the sole quantum-probability readout, with preserved implementation evidence and a readout-specific margin observation. Missing artifacts limit full reproduction.
2. **Paired computational comparisons.** We report success counts, overlaps, exact paired tests and resource accounting for candidate, forward and elapsed-time controls on a frozen timestamp population.
3. **Classical interpretation controls.** We compare four architectures under a common packet-retiming recipe to delimit what clean accuracy and one optimizer establish.

Source labels [R1]–[R10] map to preserved repository reports and code in the supplement. The supplement also retains development studies, full numerical tables and the conceptual literature comparison.

2 Related work

Raw-event attacks and constrained retiming. Surrogate-gradient learning enables optimization through spiking dynamics [1]. Event additions/removals [6] and raw-event value/support attacks [7] have different feasible sets from identity-preserving retiming. Configurable-latency attacks provide a closer temporal connection [8]. Yu et al. formalize capacity-one retiming with maximum displacement, total delay and retimed-spike budgets [2]; our packet-grid controls use a local implementation. Native timestamp retiming instead permits frame-cell collisions and uses duration-scaled radii, so the two regimes require separate interpretation.

Surrogates and temporal robustness in SNNs. Lun et al. study inference-invisible surrogate gradients and propose potential-dependent surrogates and sparse attacks [9]. Robust temporal self-ensembling [10] and sparse ANN-to-SNN conversion [11] examine training and construction choices affecting adversarial behavior. These motivate architecture-aware optimization diagnostics, with perturbation sets and training interventions distinct from timing-only evaluation.

Adversarial access and derivative-free search. PGD is a standard iterative baseline [12]. Query-limited, partial-information and label-only attacks expose different interfaces [13]. TEMP uses logits and the true label without backward gradients: it is a score-based, not label-only, attack. Random-search precedents include Square Attack [14], although its image-space proposals do not directly respect temporal feasibility. Robustness evaluation [15], obfuscated-gradient analysis [16], diverse attack ensembles [17] and

adaptive attacks [18] motivate checking the implemented pipeline beyond one optimizer.

Hybrid spiking and quantum models. Brand and Petruccione develop quantum LIF neurons [19]; Jha et al. use SNN-derived EEG features with quantum kernels [20]; SPATE investigates spike-derived temporal encodings [21]. QSNN-v3 instead combines classical Conv/LIF summaries, learned angles, a data-re-uploading circuit [22] and an affine classifier on joint probabilities. The temporal features originate in the classical frontend; the circuit does not directly implement raw-event quantum LIF dynamics.

Quantum vulnerability, encoding and guarantees. Quantum adversarial learning considers classical and quantum input manipulations [23], while high-dimensional vulnerability analysis emphasizes state-space geometry [24]. Encoding gates constrain variational expressivity [25]. Comparative quantum/classical attacks [3], randomized encodings [4] and quantum robustness guarantees [5] depend on encoding and access assumptions. Certifiably robust encodings explicitly connect classical inputs, noise and classifiers [26]. Classical autoencoder distillation of quantum manipulations is another distinct intervention [27]. Our fixed-readout margin observation lacks the additional retiming-to-state bound needed for a timestamp certificate. The supplement compares these interfaces without ranking external ASRs.

3 Models and representations

Figure 1 traces the attacked timestamp-to-logit composition; Table 4 separates the experimental regimes.

3.1 Native events and temporal-grid packets

The timestamp study uses native N-MNIST events $e_i = (x_i, y_i, \pi_i, t_i)$. Event identities, coordinates, polarity, labels and event count are immutable. Timestamps are assigned to ten equal-duration bins using the clean recording endpoints. The differentiable adapter has hard count accumulation in the forward pass and triangular interpolation weights for the backward pass. Counts are capped at 255 by the adapter. The hybrid frontend then applies $\min(X/8, 1)$. These input transformations are model preprocessing; the preservation claim refers to the underlying events.

The grid study instead freezes a ten-bin equal-event-count representation. Each nonzero cell on a fixed spatial/polarity line is one amplitude-bearing packet. A binary packet has value one; an integer packet carries its complete cell count. A packet with count five remains a single packet carrying five during retiming. Collision-free capacity-one projection prevents packet splitting, merging, deletion, creation and amplitude changes. This distinction matters because multiple raw events can occupy one cell in the timestamp study, while one grid packet is indivisible in the second study.

3.2 The implemented hybrid classifier

The Conv/LIF extractor uses two 3×3 bias-free convolutions with 16 and 32 output channels, batch normalization and average pooling. LIF dynamics use decay 0.5. Temporal information reaches the circuit through this classical frontend: time-averaged spike activations and the final membrane state are pooled to 4×4 spatial summaries and concatenated. A learned projection followed by tanh produces 32 latent features. A learned 32-to-8 projection and another tanh feed the circuit head. An additional learned 8-to-8 projection, initialized to the identity, produces bounded rotation angles.

For the latent vector $h \in \mathbb{R}^{32}$, the learned angle map is

$$\begin{aligned} u &= \tanh(W_{32}h + b_{32}), \\ a &= \pi \,\mathrm{clip}(W_8 u + b_8, -1, 1). \end{aligned} \tag{1}$$

The eight-qubit register starts in $|0\rangle^{\otimes 8}$. Each of two blocks applies an RY upload of each angle, an RZ upload scaled by a learned phase parameter, learned RY/RZ gates, and a ring of CNOT gates. Learned local RZ/RY rotations set the measurement basis. The model returns all 256 computational-basis probabilities to a linear ten-class classifier. All paths from the latent features to the logits pass through the circuit; there is no direct latent-to-logit branch.

$$\begin{aligned} q_k &= \left|\langle k|U_\theta(a)|0\rangle^{\otimes 8}\right|^2, \quad k = 0, \ldots, 255, \\ z &= W_{\mathrm{out}} q + b_{\mathrm{out}}. \end{aligned} \tag{2}$$

The backend is an analytic batched PyTorch state-vector simulation with complex64 amplitudes. Probabilities are computed exactly within numerical precision. The experiments do not involve quantum hardware, finite-shot readout or hardware noise. The name QSNN-v3 identifies this hybrid implementation and should not be interpreted as a network of physical quantum LIF neurons [R1].

Table 1. Notation used in the timestamp composition.

Symbol	Meaning
$(x_i, y_i), \pi_i, t_i$	Event coordinates, polarity and time
c, $\widehat{c}$	True class and predicted class
q, z	Measurement probabilities and logits
D, ε_f	Inclusive duration and relative timing radius
Q_{req}, Q_{act}	Requested and generated TEMP proposals

Table 2. Trainable parameters of QSNN-v3.

Component	**Parameters**
Conv/LIF frontend and 32D summary	37,792
Learned $32 \to 8$ projection	264
Angle map and quantum head	136
Probability-to-logit classifier	2,570
Total	**40,762**

The quantum head includes the learned $8 \to 8$ angle map, two upload blocks and learned measurement basis. Nontrainable BatchNorm buffers and state-vector storage are excluded [R1].

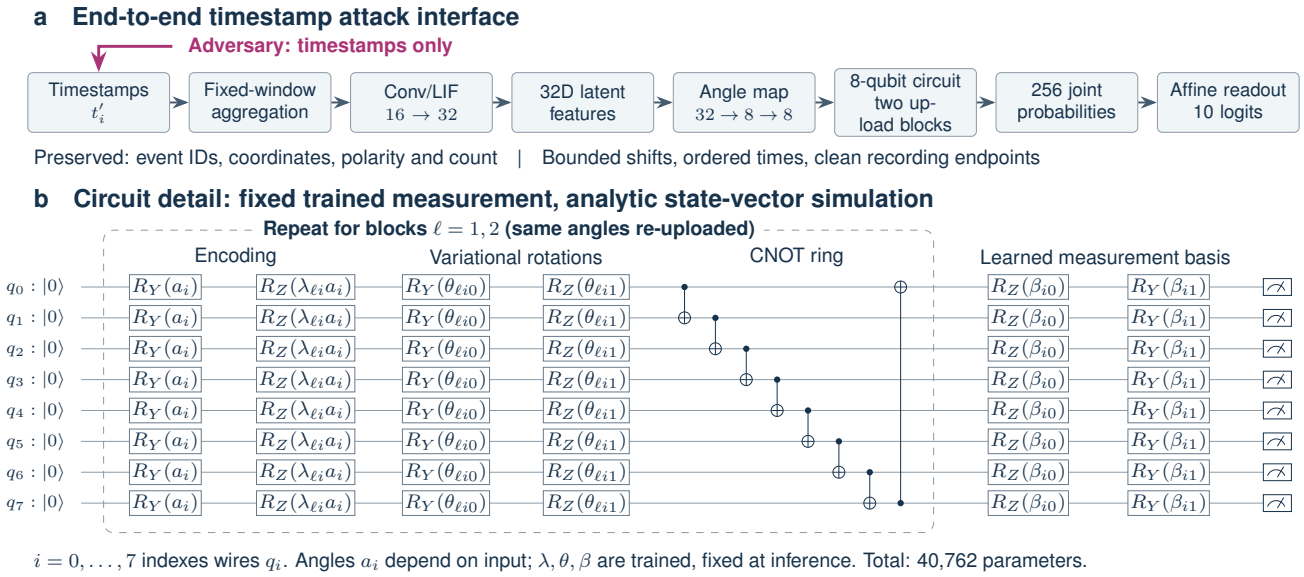


Figure 1. Timestamp attack interface and QSNN-v3 computation. Only timestamps are mutable; the aggregation window is fixed by the clean recording. Each of two circuit blocks re-uploads the angles, applies variational rotations and a CNOT ring; a learned measurement basis precedes the 256-probability affine readout. The diagram specifies computation, not an observed attack example. Source: [R1–R3].

3.3 Readout-specific margin observation

For the fixed trained affine readout $z(q) = Wq + b$, let q, q' be clean and retimed measurement probabilities. On the attacked clean-correct subset, $c = \widehat{c}(q)$ is both the true and clean predicted class; define $m_{ck}(q) = z_c(q) - z_k(q)$. With states ρ, ρ' before the fixed measurement and $D_{\mathrm{tr}} = \frac{1}{2}\|\rho - \rho'\|_1$, measurement contraction [28] and Hölder's inequality imply

$$\begin{aligned} m_{ck}(q') &\geq m_{ck}(q) \\ &- 2\|w_c - w_k\|_\infty D_{\mathrm{tr}}(\rho, \rho'). \end{aligned} \tag{3}$$

Here w_k is row k of W and the state norm is the trace norm. Strict positivity of the right-hand side for every $k \neq c$ suffices to preserve the label. The supplement gives the derivation. No uniform retiming-to-state bound or final-model per-example state measurements are available, so this is an analytical interpretation, not a certified timestamp radius.

4 Threat models and attacks

4.1 Timestamp retiming

We consider untargeted test-time evasion. For a sorted clean timestamp vector t, the feasible retiming keeps each event within its clean timestamp neighborhood and within the recording endpoints, and preserves a nondecreasing sequence. If D is the clean duration, the perturbation budget is $\varepsilon = \varepsilon_f$ D. The reported matched comparisons use $\varepsilon_f = 0.05$ and 0.10. Equal timestamps are permitted; capacity-one grid occupancy is not part of this threat model.

$$\begin{aligned} |t'_i - t_i| \leq \varepsilon_f D, \qquad & t_1 \leq t'_i \leq t_N, \\ t'_i \leq t'_{i+1}, \qquad & D = t_N - t_1 + 1. \end{aligned} \tag{4}$$

The extra unit in D follows the implementation's inclusive timestamp convention. The implementation first clips a candidate to eventwise intervals, applies a cumulative maximum, and clips to the recording range. We describe this as a feasibility operator, not as the Euclidean projection onto the constrained set. Event counts over the fixed recording window are preserved. Inter-spike intervals, time-bin counts and detector statistics can change. No completed final-model experiment establishes statistical indistinguishability from clean inputs.

Timestamp PGD. Timestamp PGD uses white-box gradients of cross-entropy through the hard-forward/soft-backward frame adapter and the surrogate-gradient LIF model. It starts at the clean timestamps, takes 20 sign-gradient steps with step size $\varepsilon/5$, and applies the feasibility operator after every step. The candidate with the highest cross-entropy among the clean input and 20 iterates is retained. There is no random start in the preserved implementation [R2]. Selection does not explicitly retain any successful intermediate candidate: a higher-cross-entropy candidate may still be correctly classified. This is a verified selection rule, not evidence that a reported success was lost; testing its empirical effect requires the missing trajectories.

$$\begin{aligned} g^{(k)} &= \operatorname{sign}\left(\nabla_t \mathrm{CE}(z(t^{(k)}), c)\right), \\ t^{(k+1)} &= \Pi_\varepsilon\left[t^{(k)} + \frac{\varepsilon}{5} g^{(k)}\right]. \end{aligned} \tag{5}$$

TEMP-DRIFT-v2. TEMP-DRIFT-v2 is a derivative-free search using model logits and the true label. It minimizes the true-class margin. A negative margin indicates that a competing logit exceeds the correct-class logit. This objective is a classification objective and contains no fidelity, density-matrix distance or classical-statistics penalty.

$$\min_{t' \in \mathscr{F}_\varepsilon} m(t'), \qquad m(t) = z_c(t) - \max_{j \neq c} z_j(t). \tag{6}$$

The preserved full-budget implementation evaluates 600 uniformly perturbed feasible candidates. It retains the 12 candidates with lowest margin, then runs four generations of

250 children. Each child combines an elite parent, half the difference between two elite candidates and Gaussian noise at scales 0.50, 0.30, 0.18 and 0.10 times ε. Every proposal is made feasible before evaluation. The final candidate with lowest margin is returned. This configuration uses 1,600 candidate evaluations [R2]. The matched implementation uses 13 initial candidates plus four generations of two children for $Q = 21$, and 13 plus four generations of seven for $Q = 41$. It always keeps 12 elites; the four noise scales are unchanged. For elapsed-time budgets it sets $n_0 = \max(12, \mathrm{round}(0.375Q))$ and $n_g = \mathrm{round}((Q - n_0)/4)$, recording the actual total $n_0 + 4n_g$. The selected $Q = 4{,}000$ and 1,600 give $1{,}500 + 4 \times 625$ and $600 + 4 \times 250$, respectively. The supplement gives the complete schedule and pseudocode [R3].

A finite search need not find more successes at a larger allowed radius; the observed 4-to-3 decrease under access matching is not evidence of a smaller feasible set. Each reported radius is a standalone search. Any future cumulative best-so-far curve must count the earlier searches and warm starts. TEMP-DRIFT-v2 is the historical implementation name; throughout the final timestamp comparison it denotes margin search.

Inference state. Both timestamp victims are loaded in evaluation mode, fixing BatchNorm statistics. Their forward methods initialize membrane and output accumulators locally on every call; the preserved code has no cross-candidate persistent neuron state. The recovered grid reference forwards also call `reset_net`; this is static evidence, not a batching/order/reset experiment on the saved victims.

4.2 Discrete packet retiming

In the second study, let s_i and s'_i denote a source packet bin and its destination. Budgets count packet displacements independently of amplitude. A packet carrying amplitude five and shifted by two bins contributes two to B_1 and one to B_0. Capacity-one projection preserves packet identity, amplitude, event line and total represented mass.

$$B_\infty = \max_i |s'_i - s_i|, \quad B_1 = \sum_i |s'_i - s_i|, \\ B_0 = \sum_i \mathbf{1}[s'_i \neq s_i]. \tag{7}$$

The local projected-in-the-loop attack uses differentiable shift probabilities and evaluates feasible hard schedules. Only B_∞ = 1 and B_0 = 200 are complete in the controlled four-model campaign [R4]. The recovered adapter selects the pinned upstream L0 class, whose loss hard-codes a normalized budget-penalty coefficient of five. Its separate capacity penalty is `lambda_cap=20`; the constructor default `lambda_B=0` does not set the hard-coded coefficient. The report compares five with the reference description of ten [R4]. This source-level finding does not reconstruct the historical runtime configuration. Historical sweeps from other checkpoints appear in the supplement.

5 Experimental protocol

5.1 Training and model selection

N-MNIST contains event recordings of handwritten digits [29]. The hybrid campaign uses a frozen stratified partition of 55,000 training and 5,000 validation examples from the official training set. Architecture selection was performed with seed 42 and validation data. The chosen wide4 frontend and project_measure2 circuit were then held fixed across model seeds 42, 123, 777, 2026 and 6543. The official test partition was not instantiated for the final QSNN campaign [R1]. Consequently, its reported clean accuracy is validation accuracy after model selection.

All five hybrid runs load the same hash-verified, seed-42 classical frontend checkpoint. The frontend is frozen for the first two epochs, then optimized with its BatchNorm layers kept in evaluation mode. The random seeds vary subsequent initialization and training stochasticity; they do not represent five independently pretrained frontends or five data splits. This conditional design limits the scope of the reported between-seed standard deviation.

Training uses AdamW with weight decay 10^{-4} and batch size 256 for at most 40 epochs. Learning rates are 5×10^{-5} for the extractor, 2.5×10^{-4} for the projection/circuit, and 10^{-3} for the classifier. A ReduceLROnPlateau scheduler uses factor 0.5, patience 2 and minimum rate 10^{-5}. The highest validation accuracy selects a checkpoint, with validation loss breaking ties; early stopping uses patience 6 after at least eight epochs. Gradients are clipped at norm 1. The frontend uses float16 autocast; circuit inputs and amplitudes use float32 and complex64. Seed 42 was externally interrupted during epoch 38; its saved epoch-37 checkpoint was re-evaluated [R1].

5.2 Populations and attack budgets

The timestamp protocol freezes 100 validation examples, with ten examples per class, correctly classified by both models before attack. It reuses their identities across conditions. This joint-correct restriction permits paired attack comparisons but excludes examples that only one model classifies correctly. It is not a population-wide robust-accuracy estimate. The compact SNN checkpoint in this study differs from the retrained custom-SNN checkpoint in the controlled grid study.

PGD evaluates 21 candidate timestamps, including the clean candidate. Its implementation makes 41 attack-internal forward passes and 20 backward evaluations because candidate scoring and gradient computation use separate forwards. TEMP scores Q proposed examples in batched model calls and uses no backwards. The first matched condition gives TEMP 21 candidates; the second matches 41 evaluated examples, not 41 batched Python model calls (Table 3). Neither converts a backward pass into an assumed number of forward passes. Verification adds two forwards per new record and is excluded from attack timing.

Elapsed-time calibration selects Q = 4,000 for QSNN and Q = 1,600 for SNN. The reported median TEMP/PGD timing ratios are 0.978 and 1.083, respectively [R3]. Candidate matching, forward matching and elapsed-time matching answer different computational questions. Exact equality of elapsed time is not asserted, and no throughput or latency claim is inferred from the ratios alone. The source syn-

Table 3. Historical search accounting per attacked example. Batched calls follow the source schedule; measured counters are unavailable.

Search	Proposals	Model calls	Backwards
PGD (20 steps)	21	41	20
TEMP, $Q = 21$	21	5	0
TEMP, $Q = 41$	41	5	0
TEMP, QSNN time	4,000	64	0
TEMP, SNN time	1,600	26	0

PGD scores 21 proposals and repeats 20 inputs for gradients (sum of forward batch sizes 41). TEMP's sum is Q; chunk size is at most 64, applied separately to initialization and each generation. Unique proposals are unknown. Two additional verification calls are excluded from historical timing. These are inferred costs, not recovered counters or equal total compute [R2–R3].

chronizes CUDA around timing, warms up PGD and TEMP, and uses ten disjoint training calibration examples (one per class), two repeats and $\varepsilon_f = 0.10$. TEMP evaluates candidate chunks of at most 64; PGD handles one example per forward. Timestamp tensors are float32 and circuit amplitudes complex64; training autocast settings are not assumed to describe attack timing. Saved timing arrays and a run-specific hardware/precision snapshot are unavailable. The supplement records calibration details and remaining provenance limits.

The controlled grid campaign trains four architectures under a common local recipe, data split, seed, optimizer settings and validation-selected checkpoint rule [R4]. For each representation it selects the first 1,000 official-test examples that all four models classify correctly. The joint-correct pools contain 9,522 binary and 9,495 integer examples. Within each representation the same frozen manifest is used for every architecture and budget. The two representations use different manifests. The public source snapshot contains the controlled-model loader, reference definitions, training and attack scripts. ConvNet denotes spiking `SimpleNet_v2`, not the separate ReLU CNN class; classical here means nonquantum. The reference source files match the pinned upstream implementation. Exact reconstruction still requires the missing resolved training configurations, checkpoints, manifests and recorded hashes. Source availability does not establish replay of every historical result. The supplement records classes, aggregation and source hashes.

The recovered grid training script selects checkpoints by validation accuracy/loss but uses final test accuracy ≥ 0.95 to mark a run PASS and decide skip/revise actions. This creates a possible test-dependent recipe gate. Logs establishing whether that branch actually changed historical recipes are absent; test independence of recipe development therefore remains unverified. Fresh campaigns must freeze the recipe on validation and report test outcomes without a test-accuracy completion threshold, retaining validity, leakage and hash gates.

5.3 Metrics, paired tests and evidence

Clean accuracy uses the complete stated partition. ASR is the fraction of the stated clean-correct attack population that becomes misclassified. Thus $1 - \text{ASR}$ is survival under the evaluated search on that selected subset, not unconditional robust accuracy. Joint-correct selection improves within-study comparability but changes the estimand relative to all examples.

For the stated clean-correct attack population $\mathscr{P}$, we use

$$\text{ASR}(\mathscr{P}) = \frac{1}{|\mathscr{P}|} \sum_{i \in \mathscr{P}} \mathbf{1}\left[f(X_i') \neq c_i\right]. \tag{8}$$

Figure 2 uses 95% Wilson score intervals [30] computed from the reported numerator and $n = 100$; the formula is given in the supplement. These are descriptive binomial-model intervals, with no sampling-based claim for the deterministically selected population or across reused conditions. They are not intervals for the paired difference. We report arithmetic means and sample standard deviations across the stated model seeds. For paired timestamp comparisons, b counts examples broken only by PGD and c counts examples broken only by TEMP. The exact two-sided McNemar framework [31] uses a Binomial$(b+c, 1/2)$ null distribution, with $p = 1$ for $b + c = 0$. It assesses observed paired discordance, not equality of all possible attack strengths. Reused examples across seeds and conditions are not treated as independent additional subjects.

All numerical results are taken from preserved reports and source files at repository snapshot `80d02f5733d4` [32]; the full commit identifier is recorded in the supplement. We checked parameter arithmetic, aggregate statistics, paired counts and exact paired p-values. Model execution and per-example bootstrap resampling were not repeated. Missing raw outputs and provenance boundaries are documented in Sec. 7 and the supplement.

6 Results

The primary result is the paired timestamp comparison on QSNN-v3. Clean validation establishes the victim's predictive performance; separate packet-grid controls delimit its interpretation. Full seed, overlap and grid tables are retained in the supplement.

6.1 Clean validation of the hybrid classifier

QSNN-v3 reaches 97.27% $\pm$ 0.21% validation accuracy and reported macro-F1 of 97.27% $\pm$ 0.21% across five training seeds [R1]. The sample SD is conditional on shared pretraining and split, as specified in Sec. 5; it is not attack-rate uncertainty. The preserved SNN validation reference reaches 98.13% $\pm$ 0.24% accuracy, a descriptive QSNN-minus-SNN difference of -0.86 percentage points. A matched validation-F1 comparison is unavailable.

The seed-42 wide4 classical frontend reaches 97.80% validation accuracy with 38,122 parameters, compared with 97.06% and 40,762 parameters for its hybrid continuation. Different training stages and budgets prevent attributing this difference to an isolated circuit component. The supplement retains individual accuracies, selected epochs and development records; no quantum accuracy advantage or official-test hybrid result is established.

6.2 Matched timestamp attacks

Each cell uses the same 100 joint clean-correct validation examples (seed 42), so one success equals one ASR percentage point. Figure 2 groups the candidate and forward conditions because their aggregate outcomes coincide; these

Table 4. Experimental populations and budget units. These regimes are analyzed separately.

Study	Models	Attack population	Budget
Native timestamps	QSNN-v3 and compact SNN	100 common clean-correct validation examples; 10/class; seed 42	$\varepsilon_f \in \{0.05, 0.10\}$ of recording duration
Controlled grid	Custom SNN, ConvNet, spiking ResNet18, VGGSNN	First 1,000 common clean-correct official-test examples per representation; seed 42	$B_\infty = 1$ bin or $B_0 = 200$ packets
Supporting studies	Earlier models/checkpoints	Study-specific subsets stated in the supplement	Study-specific timing or packet budgets

The compact timestamp SNN and retrained grid custom SNN are different checkpoints. Binary and integer grid attacks use separate manifests, drawn from joint-correct pools of 9,522 and 9,495 test examples, respectively [R2–R5].

conditions remain different resource definitions. At both radii, PGD succeeds on 3/100 examples. TEMP succeeds on 4/100 and 3/100 at 5% and 10% under either access control, versus 4/100 and 5/100 under elapsed-time calibration. The latter uses $Q = 4{,}000$ for QSNN-v3, with a reported median TEMP/PGD runtime ratio of 0.978 [R3].

The paired comparison is determined by discordant examples. At 5%, all controls have three shared successes, one TEMP-only success and 96 failures of both attacks ($p = 1$). At 10%, access matching has three shared successes and 97 failures of both ($p = 1$); elapsed-time matching has three shared, two TEMP-only and 95 failures of both ($p = 0.5$). There are no PGD-only successes in these cells. Thus two examples account for the largest numerical contrast; no matched condition establishes TEMP superiority, and nonsignificance does not prove equivalence.

For the compact SNN, candidate and forward matching produce no successes at either radius. Elapsed-time matching uses $Q = 1{,}600$ and a median runtime ratio of 1.083, producing 0/100 versus 0/100 at 5% and 0/100 PGD versus 1/100 TEMP at 10% ($p = 1$). These selected-population observations do not estimate full-set robust accuracy or general hybrid-versus-SNN robustness. The supplement preserves every matching condition, reported bootstrap differences and attributed distortion summaries.

6.3 Separate packet-grid controls

The historical grid study includes ConvNet (spiking `SimpleNet_v2`) and uses different victims, capacity-one packets and its own 1,000-example joint clean-correct test manifest for each representation (Sec. 5). Under the two completed budgets, custom-SNN ASR spans 98.0–100%, whereas the three reference architectures span 0–1.9% (Fig. 3). Clean test accuracy, measured on all 10,000 examples, is shown separately in Table 5; the supplement supplies all sixteen attack cells.

The integer custom SNN and spiking ResNet18 differ by only 0.02 percentage points in clean accuracy, yet by 99.7 and 97.9 ASR points under $B_\infty = 1$ and $B_0 = 200$. Clean accuracy therefore does not summarize the failure profile under this recipe. Common settings do not guarantee equally strong optimization across architectures, and zero observed successes are not robustness certificates [16, 17]. These are single-seed local controls, not values copied from external benchmarks or a representation ablation.

The grid report attributes passing reconstruction/prediction audits to all cells, but the underlying artifacts are absent from the supplied snapshot. The unresolved B_0 penalty discrepancy described in Sec. 4 further limits replication claims [R4].

Table 5. Clean accuracy (%) on all 10,000 official-test examples. ASR uses separate subsets.

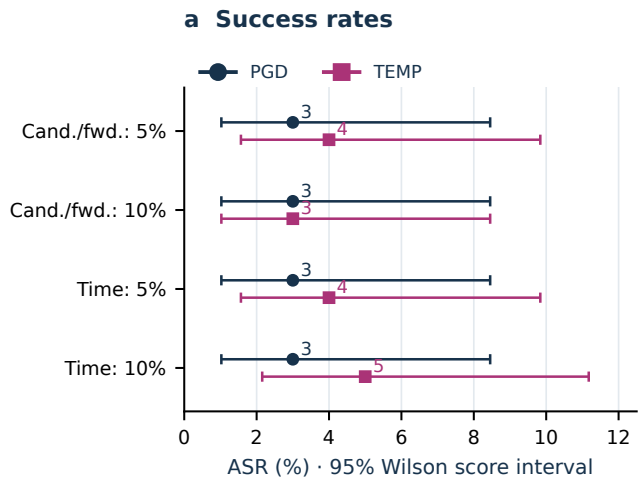
Architecture	Binary	Integer
Custom SNN	97.99	98.14
ConvNet	96.90	96.58
Spiking ResNet18	98.61	98.12
VGGSNN	99.21	99.30

7 Discussion and limitations

The primary timestamp experiment identifies feasible retimings that change QSNN-v3 predictions. Its numerical PGD/TEMP contrast changes with computational matching: the two extra failures at the larger radius occur under elapsed-time calibration and disappear under access matching. The paired tests remain inconclusive. The classical controls answer a separate question: similar clean accuracy need not imply similar outcomes under the same nominal attack recipe. Different victims, samples and budget units prevent pooling the two studies.

Mechanism and encoding. The quantum circuit is one stage of an attacked composition. Timestamp changes can affect bin counts, spiking summaries, angles and measurement probabilities before the affine readout. Without fixed-frontend classical-head and circuit ablations, failures cannot be attributed uniquely to quantum processing. The margin observation explains why state displacement alone need not change a label, but final-model per-example states and margins are missing. Historical Iris diagnostics in the supplement do not supply that evidence. Encoding- and access-specific quantum analyses [3–5] consequently provide context rather than guarantees for our temporal feasible set.

Search strength. Low or zero ASR can reflect optimizer limitations. PGD uses one start and 20 steps, with cross-entropy-based output selection; TEMP uses a finite margin search. The large classical architecture gap requires gradient, preprocessing and search-strength diagnostics before interpretation as an intrinsic robustness difference. Alternative surrogates, restarts and derivative-free budget curves should be checked with valid hard-forward candidates [16, 17]. Image-space attack ensembles and defenses, including classical distillation of quantum manipulations [27], require adaptation to event constraints.



b Paired counts (n=100 per row)

Both	PGD-only	TEMP-only	Neither
3	0	**1**	96
3	0	0	97
3	0	**1**	96
3	0	**2**	95

Figure 2. QSNN-v3 timestamp outcomes on the same $n = 100$ joint clean-correct validation examples, seed 42. (a) ASR with descriptive 95% Wilson score intervals; numbers beside markers are success counts. Candidate ($Q = 21$) and forward ($Q = 41$) matching share the displayed aggregate counts. (b) Exact paired counts, including the small discordant groups. Equal aggregate rows for candidate and forward matching do not establish the same failed IDs across these conditions. Intervals are binomial-model summaries for this selected subset, not independent evidence or uncertainty over all validation/test examples. Source: [R3].

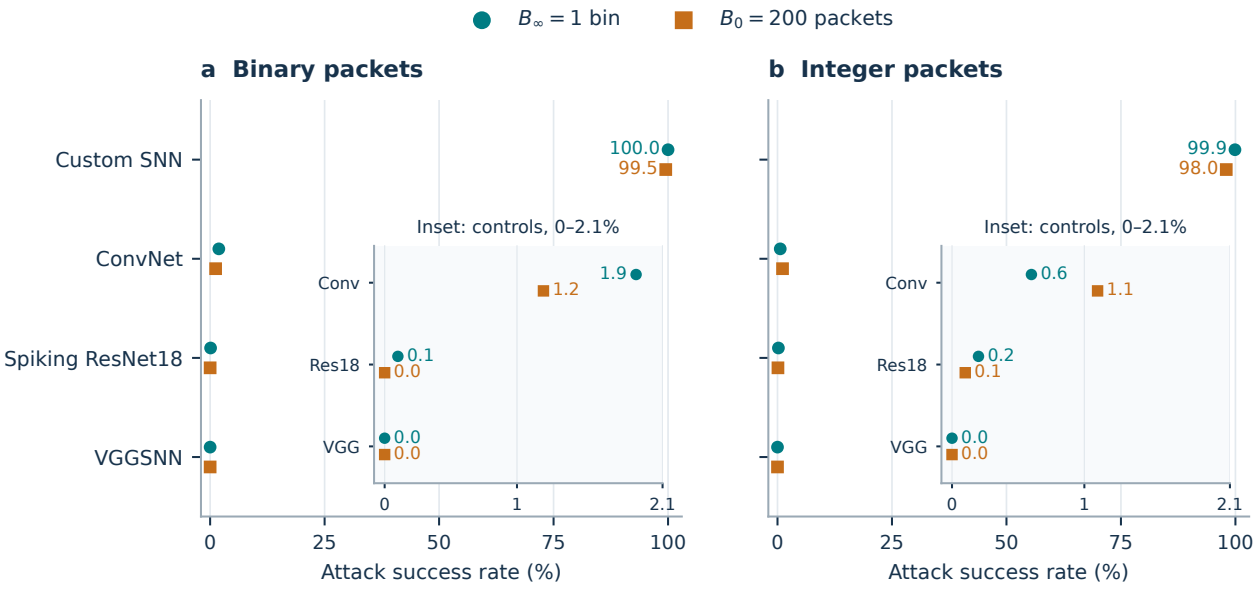


Figure 3. Historical nonquantum controls; ConvNet is spiking `SimpleNet_v2`. Maximum displacement ($B_\infty = 1$ bin) and retimed-packet count ($B_0 = 200$). Each cell uses $n = 1{,}000$ joint clean-correct test examples, seed 42. Insets explicitly expand 0–2.1% for the three low-ASR controls. Binary and integer panels use different manifests; the panels are not paired representation comparisons. Source: [R4].

Evaluation and evidence boundary. Five hybrid seeds share pretraining and split, while attacks use one victim seed and a selected validation population. Official-test QSNN robustness, independently pretrained pipelines and a larger attack population remain unmeasured. The populated comparison report conflicts with its README's statement that results are absent; we use the report and disclose that inconsistency. Most checkpoints, raw arrays, manifests and logs referenced by reports are missing. Our arithmetic and static code checks do not reproduce model runs or independently repeat their attributed audits. Recovered baseline source and the upstream fixed penalty clarify implementation provenance; resolved runtime settings and their historical linkage remain incomplete. The test-dependent training gate is an additional limitation: validation-selected weights alone do not establish that recipe development was independent of test outcomes.

Next empirical steps. After validation-based model selection, evaluate QSNN clean and attacked performance on the official test set across independently trained frontend/head seeds, retaining sample IDs and clean errors. Sweep steps, step size, restarts, cross-entropy/margin losses and surrogate choices, recording the best valid success by budget and reconfirming it with hard forwards. Hold frontend, split and training budget fixed for capacity-matched classical heads and circuit variants with/without entanglement, including parameter counts and training cost. To test the state/margin interpretation, retain per-example states, probabilities and

margins and control clean margin and realized distortion; mean fidelity alone would not establish a mechanism.

A future monitor must specify eight-qubit observables, shot budget, clean-only calibration and response policy before measuring ROC/PR, false-positive rates and adaptive attacks. Single-qubit marginals do not determine the full state. Event-count preservation is also distinct from stealth: no completed statistical-indistinguishability or final-model detection/defense evaluation exists. The historical corrected grid campaign remains only 8 of 33 conditions complete.

8 Conclusion

Computational matching changes the numerical attack comparison on the implemented QSNN-v3, but none of the paired conditions establishes TEMP superiority. Constrained retiming changes 3–5 of 100 examined joint-correct validation predictions. Separate classical controls show why clean accuracy and a common recipe are insufficient summaries of attack behavior. The resulting evidence is specific to the stated protocols; independent official-test hybrid evaluation is the principal missing validation.

Data and code availability

The `ABC_def1` repository [32] and supplied snapshots contain model/attack/training source and preserved reports. Source hashes and a source/config/weights/data/results/audit availability matrix accompany the supplement and manuscript revision report. Resolved historical configs, weights, raw data, per-example results and run-audit artifacts are missing locally. Tables transcribe reports; figures reorganize their counts. Reproduction instructions specify the missing prerequisites and portable software-only checks. Author details remain withheld for review.

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